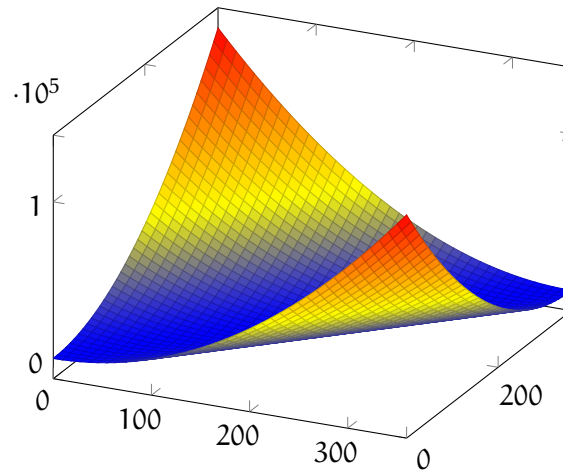
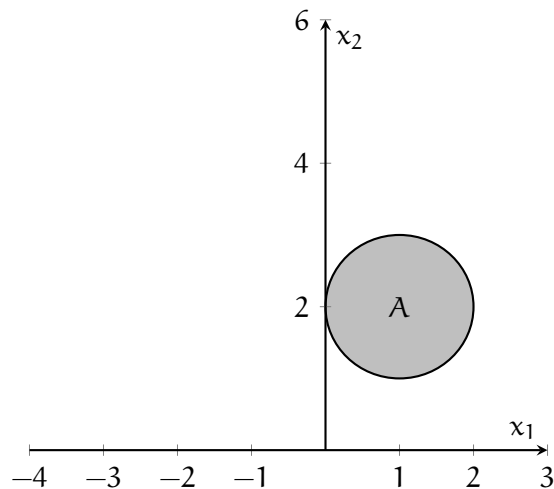


The function $f(x_1, x_2) = x_1^2 + x_2^2 - 2x_1x_2$ is a continuous function, because it is the sum of continuous functions.



The set $A = \{(x_1, x_2) \in \mathbb{R}^2 | (x_1 - 1)^2 + (x_2 - 2)^2 \leq 1\}$ is a compact set:



Weierstrass's theorem guarantees that the given function has both a maximum and a minimum when optimized on the feasible set A .